



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 13

Assignment: Multilayer Systems and Architectural Design

PROGRAMME

BSc Information Communication and Technology

COURSE CODE / YEAR

BICT 3202 · Year 3

LECTURER

Francis Fweta — ICT Lecturer

DELIVERY

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 13 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 13 module and your lecture notes before attempting each question.
- Use the university course registration case study for your examples.
- Component, package and deployment diagrams must use correct UML notation.
- Where you critique an architecture, propose a concrete improvement.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define software architecture and distinguish it from class design. [3 marks]

Question A2

Define a software layer and list the layers of a common multilayer architecture. [3 marks]

Question A3

Explain the responsibilities of the Presentation and Domain layers. [2 marks]

Question A4

Define a reusable component and give one example. [2 marks]

Question A5

Explain how multilayer architecture supports separation of concerns. [2 marks]

Question A6

What is a component diagram and when should it be used? [2 marks]

Question A7

Define a package and explain its purpose in UML. [2 marks]

Question A8

Explain the relationship between components and packages. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider a university management system with Presentation, Business Logic and Data layers.

- (a) Explain what belongs in each layer. [4 marks]
- (b) Draw a component diagram showing the layers. [4 marks]
- (c) Explain how this architecture supports reuse. [4 marks]

Question B2 (8 marks)

Consider the Registration component.

- (a) Identify reusable components within Registration. [4 marks]
- (b) Explain how these components could be reused in another system. [4 marks]

Section C — Case Study (10 marks)

Consider a hospital management system with multiple layers and reusable components.

As an object-oriented architect:

- (a) Design a multilayer architecture for the system. [4 marks]
- (b) Identify reusable components and explain their value. [3 marks]
- (c) Explain how this architecture supports maintainability and scalability. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — hospital system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Diagrams must use correct UML notation for components and packages.
- Cite any sources you consult using the format used in the Week 13 module references.